

## COMP.2560: System Programming

### LAB #5: Process Control – `fork()`

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In this lab you will familiarize yourselves with Process Control, especially with the `fork()` call.

#### LAB 5: Overview

Watch the **video** posted in Brightspace<sup>1</sup> which explains how the “`fork(...)`” function works in some sample code “`q1.c`” (briefly discussed in class). Then answer the following questions:

- Part I: Explain how `q2.c` works.
- Part II: Write a C program such that a parent process generates exactly three (3) child processes.

These two parts of the lab are described below.

#### LAB 5 – Part I: `q2.c`

The code in example “`q2.c`” was briefly discussed in class. However, we did not run/explain its output.

Write a short document or, even better, record a short video (like the one included in the Lab) to explain the output of “`q2.c`” in a similar fashion as in my video.

#### LAB 5 – Part II: Write a C program

Write a C program such that the parent process generates exactly 3 child processes. Each child process:

- Prints its pid and its parent's pid to the terminal, and
- Writes what prints to the terminal to a file, shared by all the processes.

Please put extended comments in your code so that markers can understand your code. You could also make use of the `sleep(...)` function, if necessary, to make sure the child process terminates before the parent process.

**Hint:** If you fully understand how “`q2.c`”, you will realize you only need to add one line of code to “`q2.c`” to create exactly 3 child processes.

Also, take a look at the sample code “`fork3.c`” discussed in class to see how different processes write to the same file (you may consider using the `dprintf(...)` function for writing to a file descriptor).

The screenshot below shows a sample output.

<sup>1</sup> © Dr. Dan Wu



```
danwu@delta:~/comp2560w2024/labcode$ cc lab5q2.c -o lab5q2
danwu@delta:~/comp2560w2024/labcode$ ./lab5q2
before fork, my pid is 34885
Hi, I am child. My pid is 34886, myppid=34885
Hi, I am child. My pid is 34887, myppid=34885
Hi, I am child. My pid is 34889, myppid=34885
danwu@delta:~/comp2560w2024/labcode$ cat lab5q2.txt
Hi, I am child. My pid is 34886, myppid=34885
Hi, I am child. My pid is 34887, myppid=34885
Hi, I am child. My pid is 34889, myppid=34885
danwu@delta:~/comp2560w2024/labcode$
```

#### LAB 5 – Submission

Submit the items elements described above.

To formally submit your work for this Lab:

- Submit a document file, or a short video, briefly explaining how example q2 . c works.
- Submit the code you created for Part II.

When submitting, please indicate in the submission textbox the Lab section you belong to.

#### Evaluation:

1. It is our intention to evaluate your work being done in-person, while you are in the lab.
2. It is strongly suggested that you attend in-person to familiarize yourselves with the Lab infrastructure, meet with the GATAs and solve problems.
3. Each session is only 80 minutes long, so it is strongly suggested that you arrive early, are focused and do not disrupt the process.
4. It is desired that you complete your work while in the lab and get graded after live review from the GATAs.
5. Should you need more time, you can take your work home and present it to the GATAs on the very next lab session (the week following).
6. Even if you decide to work from home and submit your work remotely, you still need to have your work peer reviewed, in-person.